

SPS_Data607_Week10B_DC

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Assignment 10B: More JSON Practice

The Nobel Prize organization provides public APIs for accessing structured Nobel Prize data. Using one or both of the APIs available at the Nobel Prize Developer Zone, your task is to **use JSON data to investigate and answer four interesting, data-driven questions**.

At least one of your questions should go beyond simple counts and require **joining, filtering, or comparing fields** across the data (e.g., “*Which country lost the most Nobel laureates—born there but awarded as a citizen of another country?*”).

Requirements

1. Use one or both Nobel Prize APIs to retrieve data in **JSON** format.
2. In **R**, load and transform the JSON data into tidy data frames.
3. Formulate **four questions** that can be answered from the data.
4. For each question:
 - Describe the question
 - Show the code used
 - Present the answer (table, summary, or plot)

Deliverables

Submit a single Quarto (`.qmd`) file that includes:

- Your four questions
- All R code used to retrieve and process the data
- The resulting answers

You may complete this assignment **individually or in a small group**.

The Nobel Prize API link

<https://api.nobelprize.org/2.1/laureates>

<https://api.nobelprize.org/2.1/nobelPrizes>

```
library(httr)
library(jsonlite)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)

if(all(file.exists(c("df_laureates.json", "df_nobel.json")))) {
  # load both files
  print("Cache files exists\n")

  json1 <- readLines("df_laureates.json", warn = FALSE)
  json2 <- readLines("df_nobel.json", warn = FALSE)
}else{
  # load from API
  #url1 <- "https://api.nobelprize.org/2.1/laureates"
  url1 <- "https://api.nobelprize.org/2.1/laureates?limit=1018"
  #url2 <- "https://api.nobelprize.org/2.1/nobelPrizes"
  url2 <- "https://api.nobelprize.org/2.1/nobelPrizes?limit=682"
  res1 <- GET(url1)
```

```

res2 <- GET(url2)

json1 <- content(res1, as = "text", encoding = "UTF-8")
json2 <- content(res2, as = "text", encoding = "UTF-8")
writeLines(json1, "df_laureates.json")
writeLines(json2, "df_nobel.json")

}

```

```
[1] "Cache files exists\n"
```

```

# identical(df_laureates, df_laureates_0)
# identical(df_nobel, df_nobel_0)
data1 <- fromJSON(json1, flatten = TRUE)
data2 <- fromJSON(json2, flatten = TRUE)
js_laureates <- data1$laureates
js_nobel <- data2$nobelPrizes

```

```

library(tidyr)
df_nobelprizes <- js_nobel %>%
  select(awardYear, laureates, category.en) %>%
  unnest(laureates) %>%
  transmute(
    l_id = id,
    laureate_name = fullName.en,
    award_year = awardYear,
    category = category.en
  )
#df_nobelprizes

```

```

df_laureates <- js_laureates %>%
  select(id, gender, 'birth.year', 'birth.place.city.en', 'birth.place.countryNow.en', 'birth.place.continent.en')

  transmute(
    l_id=id,
    gender=gender,
    birth_year=birth.year,
    city=birth.place.city.en,
    birth_country=birth.place.countryNow.en,
    birth_continent=birth.place.continent.en
  )

```

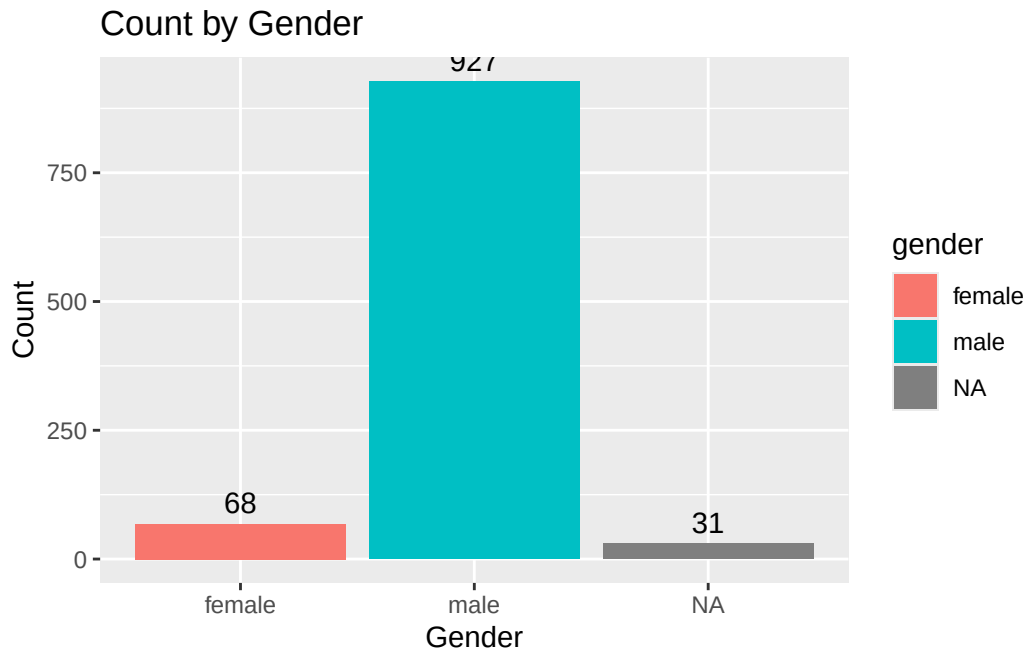
```
)  
#df_laureates  
#names(df_laureates)
```

Combine both data frames

```
df_combined_nobel <- left_join(df_nobelprizes,df_laureates,by ="l_id")  
#df_combined_nobel
```

Q1: Comparing in gender

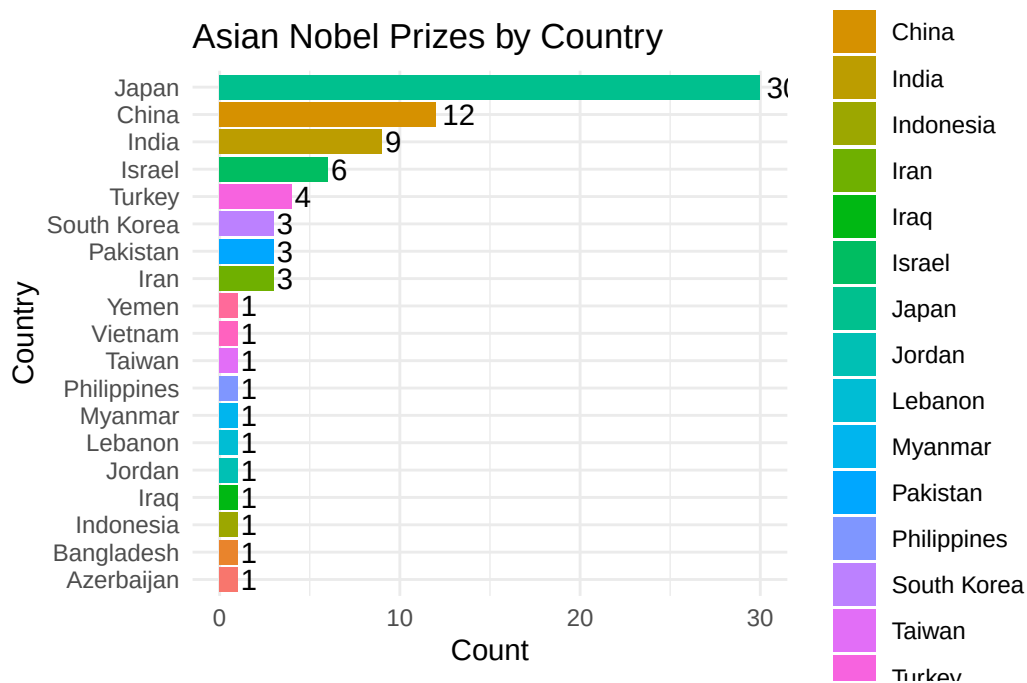
```
library(ggplot2)  
ggplot(df_combined_nobel, aes(x = gender,fill = gender)) +  
  geom_bar() +  
  geom_text(stat = "count",  
            aes(label = after_stat(count)),  
            vjust = -0.5) +  
  labs(title = "Count by Gender",  
        x = "Gender",  
        y = "Count")
```



Q2: How many Asians were awarded, and how are they distributed by country?

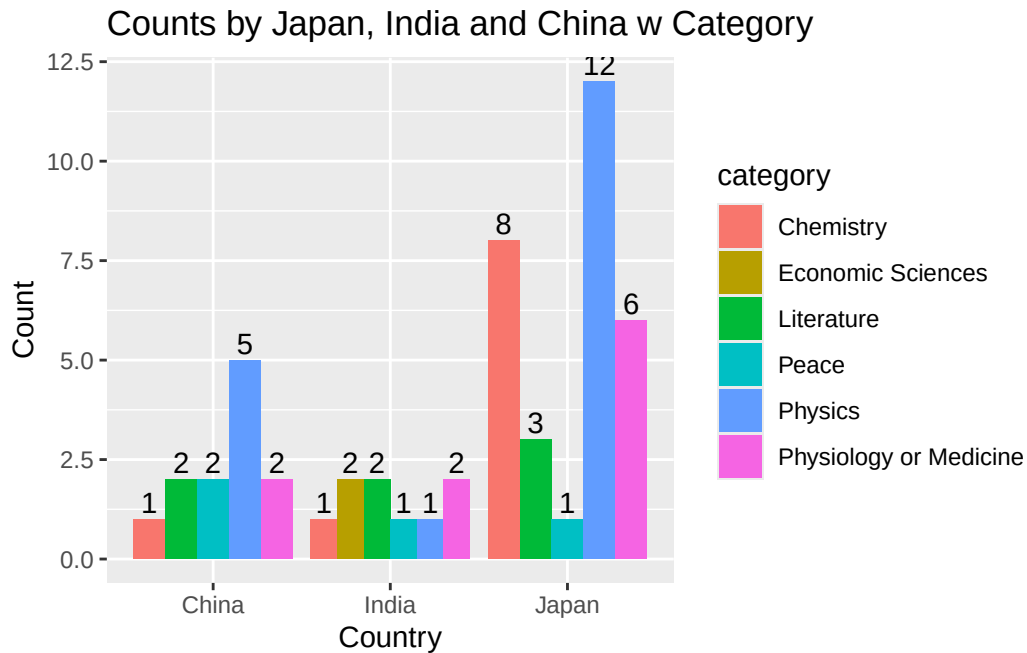
```
df_combined_nobel %>%
  filter(birth_continent == "Asia") %>%
  count(birth_country, sort = TRUE) %>%

  ggplot(aes(x = reorder(birth_country,n), y=n,fill = birth_country)) +
  geom_col() +
  # geom_text(stat = "count",
  #           aes(label = after_stat(count)),
  #           vjust = -0.5) +
  geom_text(
    aes(label = n),
    hjust = -0.2) +
  coord_flip() +
  labs(title = "Asian Nobel Prizes by Country",
       x = "Country",
       y = "Count") +
  theme_minimal()
```



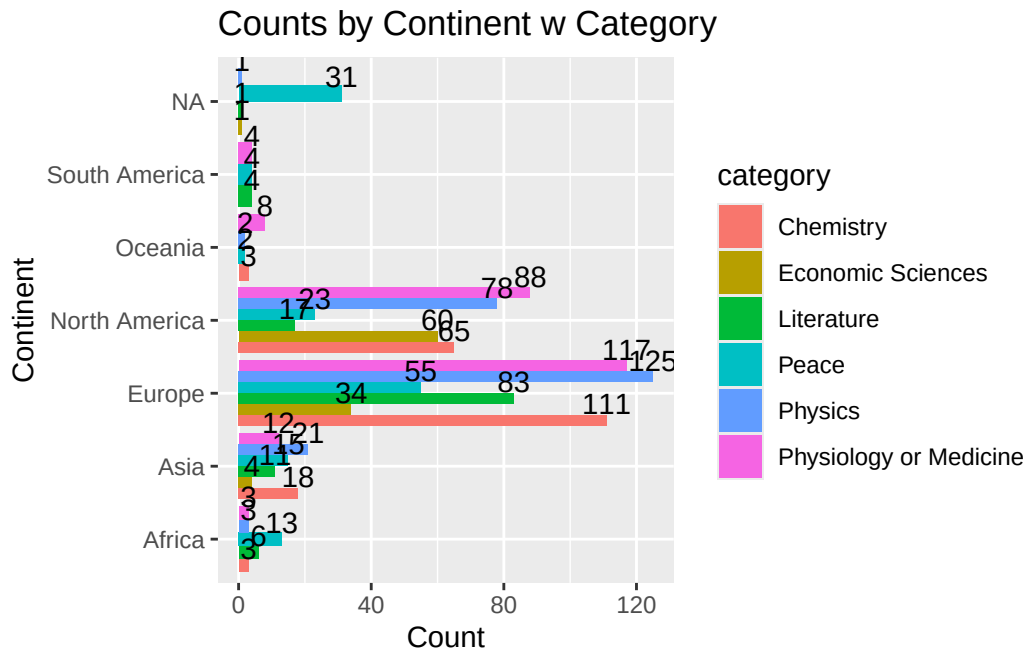
Q3 Which categories were Japan, India, and China awarded in?

```
df_combined_nobel %>%
  filter(birth_country %in% c("Japan", "China", "India")) %>%
  count(birth_country, category) %>%
  ggplot(aes(x = birth_country, y = n, fill = category)) +
  geom_col(position = "dodge") +
  geom_text(aes(label = n,
                position = position_dodge(width = 0.9),
                vjust = -0.3)) +
  labs(title = "Counts by Japan, India and China w Category",
       x = "Country",
       y = "Count")
```



Q4: Comparing the distribution of Nobel Prizes across different continents.

```
df_combined_nobel %>%
  #filter(birth_country %in% c("Japan", "China", "India")) %>%
  count(birth_continent, category) %>%
  ggplot(aes(x = birth_continent, y = n, fill = category)) +
  geom_col(position = "dodge") +
  coord_flip()+
  geom_text(aes(label = n),
            position = position_dodge(width = 0.9),
            vjust = -0.3) +
  labs(title = "Counts by Continent w Category",
       x = "Continent",
       y = "Count")
```



Conclusion

By using the Nobel Prize API, this project also provided hands-on experience in working with JSON data, including retrieving, parsing, and converting API responses into structured data frames in R. That explored Nobel Prize data using R to uncover patterns in awards across categories, countries, and time periods. Through data cleaning, transformation, and visualization, we identified clear trends in how Nobel Prizes are distributed globally.

LLMS used:

- OpenAI. (2025). ChatGPT (Version 5.2) [Large language model]. <https://chat.openai.com>. Accessed Apr 19, 2026.